



Hello ~~Hackers~~ Wildcats

Level 0x00: pwn.college



Quick Overview

- About me
- Club Agenda
- CTF
- Pwn.college

About me - Mike Wales

- Working at Research Innovations 1 year
- Work at Nightwing CODEX for 10 years
- Worked at L3Harris for 14 years
- BS Computer Engineering - UCF 2001
- Software Development
 - Mostly embedded (VxWorks, Linux)
 - Mostly C/C++
- Cybersecurity Engineer
 - Hacking
 - Reverse Engineering
 - Vulnerability Research
 - Software Developer
- 5th year of helping with CS Club
- Retro-gaming



NIGHTWING



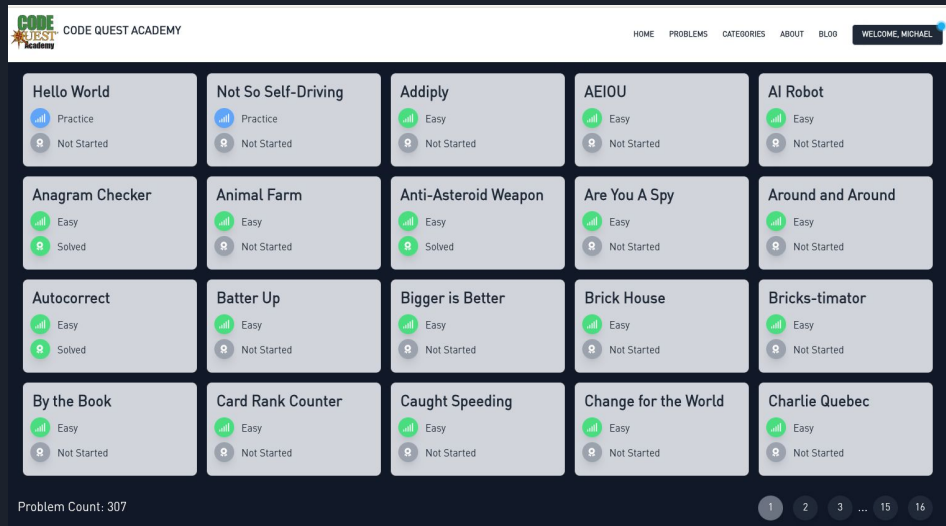


Objectives / Goals

- Computer Science Enrichment
 - Learn some things not in regular CS curriculum
- Contests / CTF
 - Lockheed Martin Code Quest - Programming
 - Lockheed Martin Cyber Quest - Cybersecurity
- Offensive Cyber Security
 - You will not find anything like this in high school curriculum

Code Quest Academy

- Programming Challenges
- Languages (your choice)
 - C/C++
 - Java
 - Python
 - C#
- 300+ challenges
 - Different difficulty levels
 - Hard can be very challenging
- From past competitions

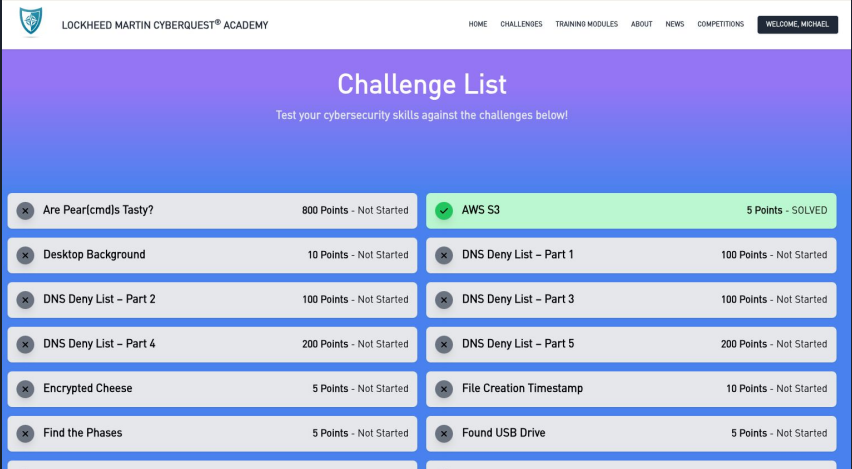


Great place for new students to start

Cyber Quest Academy

- Cybersecurity Challenges
 - Blue team / defensive
 - Incident response
- 100+ challenges

Variety of stuff to explore



The screenshot shows the 'Challenge List' page of the Lockheed Martin CyberQuest Academy. The page has a purple header with the title 'Challenge List' and a subtitle 'Test your cybersecurity skills against the challenges below!'. A navigation bar at the top includes links for HOME, CHALLENGES, TRAINING MODULES, ABOUT, NEWS, and COMPETITIONS, along with a 'WELCOME, MICHAEL' button. The challenges are listed in a two-column table, each with a status icon (X for 'Not Started', checkmark for 'SOLVED'), the challenge name, and the points and status.

Challenge Name	Points	Status
Are Pear(cmd)s Tasty?	800 Points	Not Started
Desktop Background	10 Points	Not Started
DNS Deny List - Part 2	100 Points	Not Started
DNS Deny List - Part 4	200 Points	Not Started
Encrypted Cheese	5 Points	Not Started
Find the Phases	5 Points	Not Started
AWS S3	5 Points	SOLVED
DNS Deny List - Part 1	100 Points	Not Started
DNS Deny List - Part 3	100 Points	Not Started
DNS Deny List - Part 5	200 Points	Not Started
File Creation Timestamp	10 Points	Not Started
Found USB Drive	5 Points	Not Started

Quest Contests at Orlando Lockheed



Quest Contests at Orlando Lockheed



Quest Contests at Orlando Lockheed



Quest Contests at Orlando Lockheed





Lockheed Code / Cyber Quest

- Free to sign up for (same login works with both sites)
- Not teaching platforms / just evaluation
- No feedback
 - Output for coding challenges must be exact
 - No hints / descriptions of errors



AP Cybersecurity

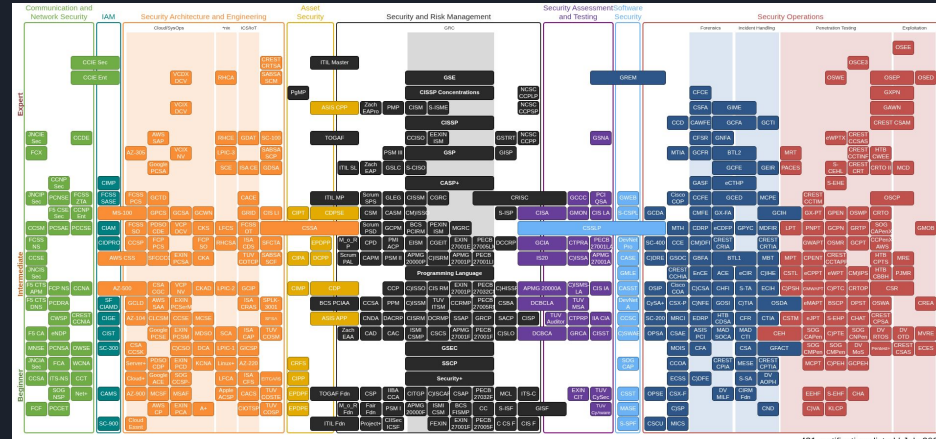
- Skills (per AP)
 - Analyze Risk
 - Mitigate Risk
 - Detect Attacks
 - Collaborate
- Units
 - Intro to Security (passwords, MFA, HTTPS, VPN)
 - Securing Spaces (phases of cyber attack, physical vulns, physical attacks)
 - Securing Networks (wifi attacks, firewall, network logs)
 - Securing Devices (hashes, detection software, detecting attacks)
 - Securing Applications and Data (data encryption, sw exploits, input sanitization)

My takeaways:

- Hands on demonstrating how to attack security vulnerability is the best way to understand many of these concepts

Cybersecurity Certificate Situation

- Cybersecurity field very diverse
- So many certificates
 - Collecting like Pokemon is red flag for a paper engineer
 - Many are expensive
 - Big industry / cash grab
- You are not an expert if...
 - Take class for a few hours / days
 - Take an exam
 - Learn a bunch of terms
- Why get them?
 - Passes HR filter



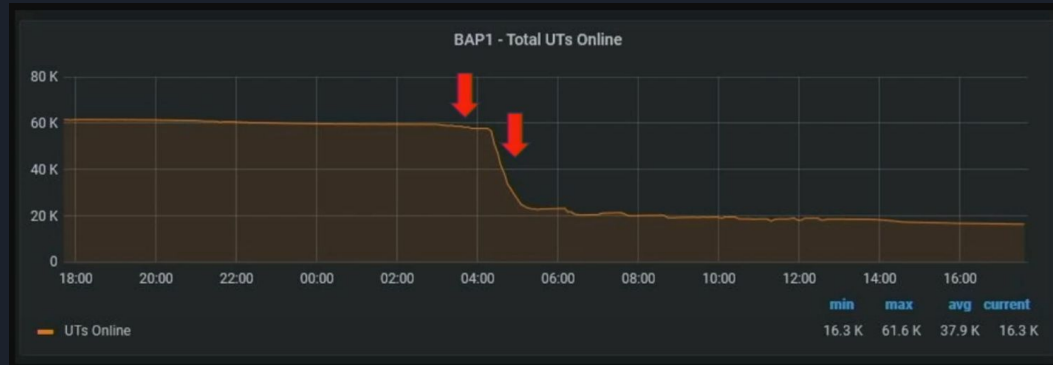
Offensive Cyber Security

- Positions / Job Titles
 - CNO (Computer Network Operations) Developer
 - Vulnerability Researcher
 - Reverse Engineer
- Skills
 - C, C++, Python
 - OS Internals
 - Linux, Embedded, Windows, Mobile (Android, iOS)
 - Networking, RF, routing
 - Assembly (x86, x64, ARM, PPC, MIPS, SPARC, 68k, or others)
 - RE and Debugging Tools: GDB, IDA, Ghidra, Binary Ninja



Why are these skills important?

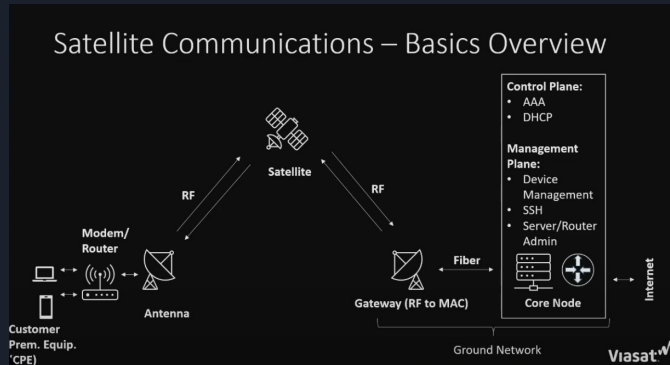
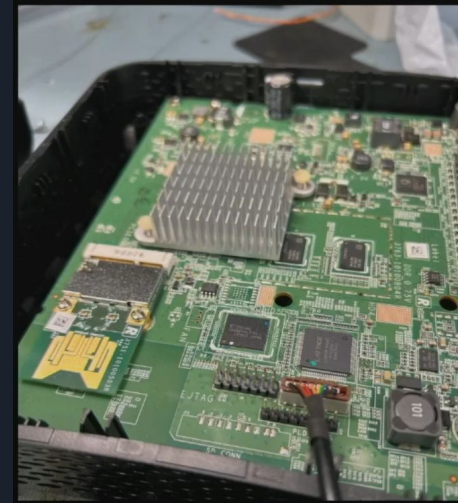
- Viasat (satellite communication) online terminals on Feb 24, 2022



- Who are the users of satellite communication terminals?

Everything bricked...

- Firmware wiped off devices
- What happened?
- How did they get in?
- How can we prevent in the future?
- How can we fix all these?





Why learn offensive security

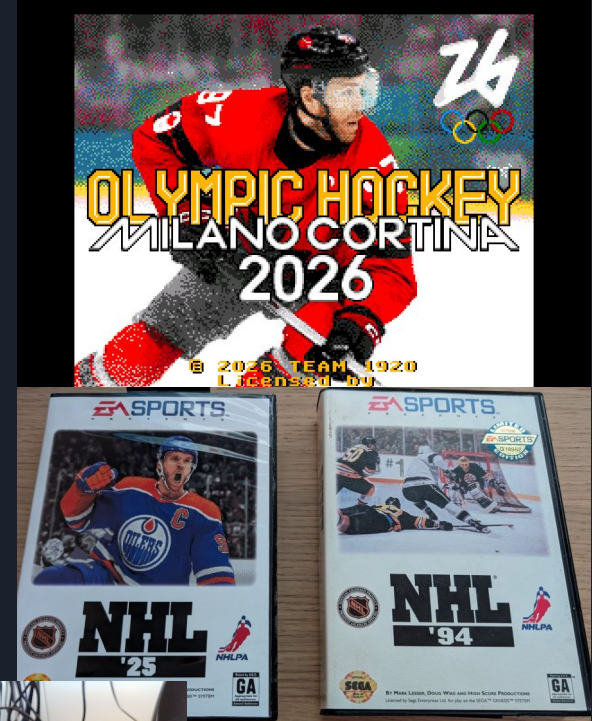
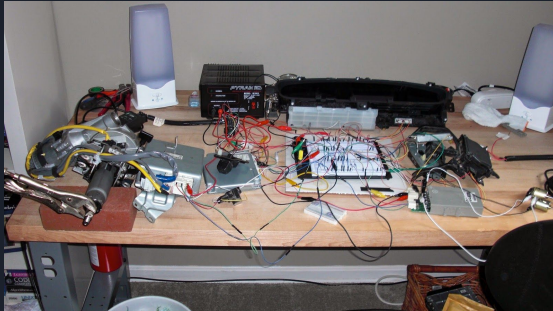
- Learning programming and scripting
- Learning how operating systems work and access controls
- Learn new software introspection techniques (debugging, tracing, emulation)
- Learn how code gets converted to executable code
 - What exactly in the file I'm executing?
 - How are symbols found?
 - How do function calls work?
 - Really understand C++ and vtables!
- Learning how vulnerabilities work gives you a foundation to avoid flaws in development
 - Why does a stack buffer overflow break / make a vulnerability
 - How can file access permissions be abused by attackers
 - How are cryptography failures exploited

More things you can do...

Hacking is just kinda fun

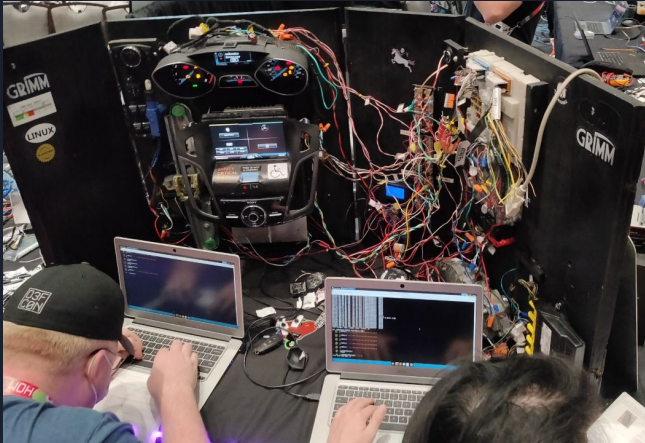
- Game hacking / ROM hacking
- Reverse engineering software
- Software modding
- Fixing broken software / devices

NEVER STOP LEARNING!



Capture the Flag (CTF)

- Hacking contest online or in-person (at security conferences)
- Attackers try to capture flag (secrets)
 - Via bug / flaw in organizers challenge
 - Forensic research
 - From other teams when playing attack and defend





- Platform by former CTF players and ASU students (now professors)
- Heavily modified version of CTFd
- Part of ASU CS required curriculum

The Courses — Earning Credit

We leverage the above material to run a number of courses on this platform. For the most part, these courses import the above material, though some might introduce new concepts and challenges.

CSE 365 - Fall
2025

1 Module
149 Challenges

CSE 466 - Fall
2025

11 Modules
267 Challenges

CSE 539 - Spring
2025

7 Modules
44 Challenges

CSE 598 AVR - Fall
2024

7 Modules
62 Challenges

pwn.college | 🏠 Dojos 🖥️ Workspace ? Help 💬 Chat 🔍 Search



Getting Started — Learn the Basics!

These first few dojos are designed to help you Get Started with the platform. Start here before venturing onwards!

Start Here



2 Modules
11 Challenges

Linux Luminarium



16 Modules
126 Challenges

Computing 101



7 Modules
69 Challenges

Playing With
Programs



4 Modules
116 Challenges

After completing the dojos above, dive into the Core Material below!

Core Material — Earn Your Belts!

These dojos form the official pwn.college curriculum, taking you on a curated journey through the art of hacking. As you progress and build your skills, like in a martial art, you will earn **belts** for completing dojo after dojo. We won't stop you from jumping around if you want (and have the requisite skills), but you must earn belts sequentially.

Intro to
Cybersecurity



7 Modules
183 Challenges

Program Security



7 Modules
188 Challenges

System Security



6 Modules
93 Challenges

Software
Exploitation



6 Modules
103 Challenges

After completing the dojos above, not only will you be added to the **belts** page, but *we will send you actual pwn.college-embroidered belts*!

To get your belt, **send us an email** from the email address associated with your pwn.college account. We'll then get your belt over to you (eventually)! Note that, due to logistical challenges, we're currently only *shipping* belts to hackers after they earn their blue belt. Until then, we will belt you in person, at ASU or some security conference.

How to enroll

- Sign up
 - Username / Hacker Handle
 - Email Address
 - Integrity Pledge
- Earn your white belt!



Start Here



2 Modules
11 Challenges

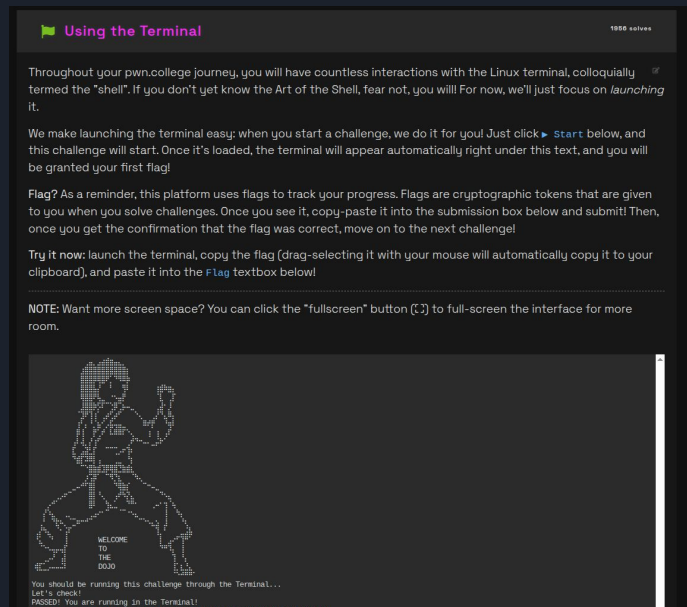
Linux Luminarium



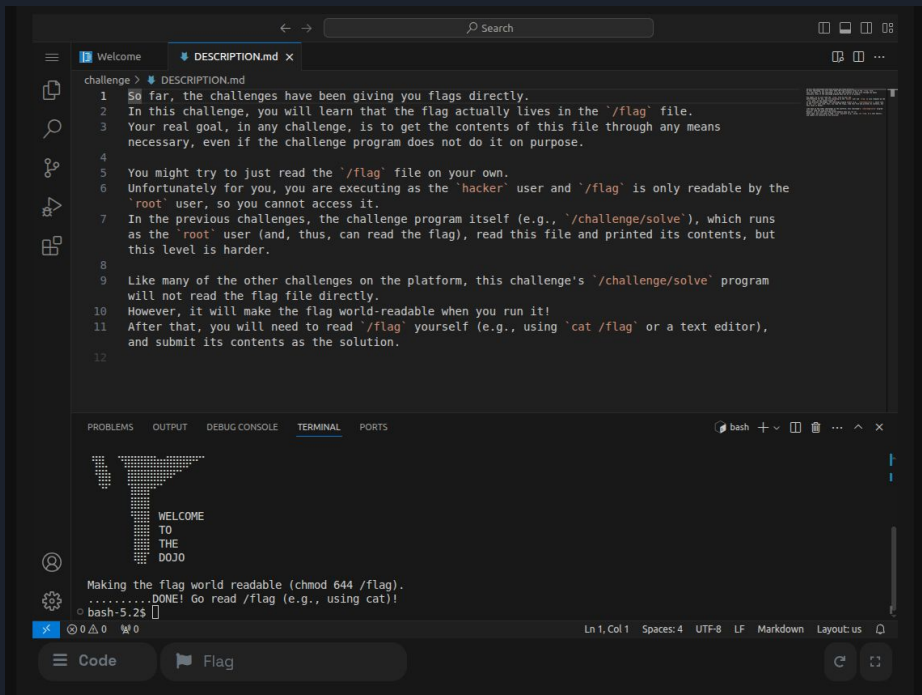
16 Modules
126 Challenges

Welcome to the Dojo

- Each dojo has submodules on different subjects
- Module contents
 - Youtube lecture
 - Google slides
 - Challenges
- Start the challenge
 - Web-terminal
 - VS Code
 - Desktop
 - SSH mode



Other options



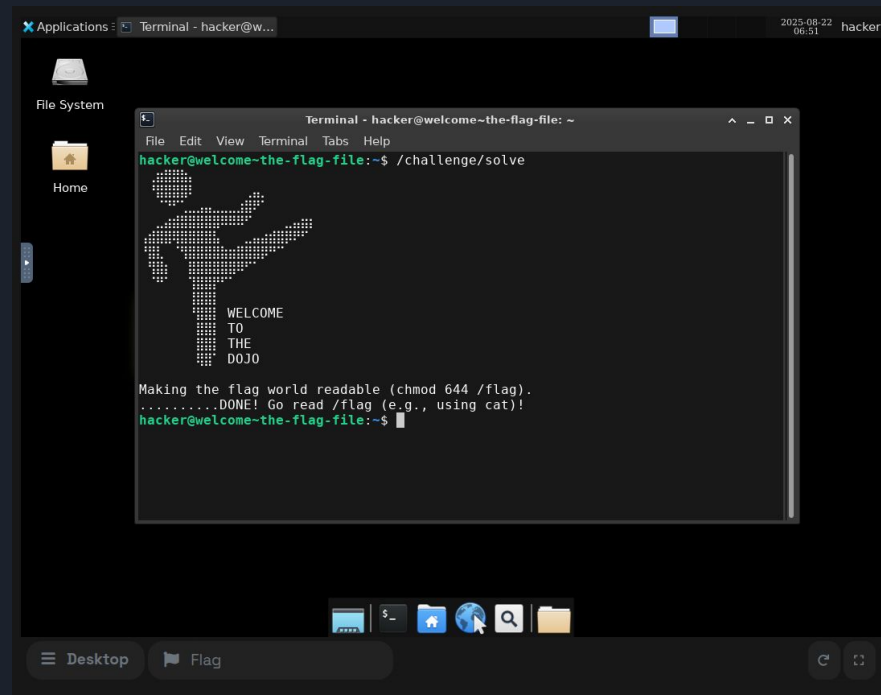
The screenshot shows a code editor with a file named `DESCRIPTION.md` open. The file contains a challenge description with 12 lines of text. Below the editor, there is a terminal window. The terminal shows a ASCII art of a person standing, followed by the text "WELCOME TO THE DOJO". Below this, it says "Making the flag world readable (chmod 644 /flag).DONE! Go read /flag (e.g., using cat)!" and then "bash-5.2\$".

```
challenge > DESCRIPTION.md
1 50 far, the challenges have been giving you flags directly.
2 In this challenge, you will learn that the flag actually lives in the '/flag' file.
3 Your real goal, in any challenge, is to get the contents of this file through any means
4 necessary, even if the challenge program does not do it on purpose.
5
6 You might try to just read the '/flag' file on your own.
7 Unfortunately for you, you are executing as the 'hacker' user and '/flag' is only readable by the
8 'root' user, so you cannot access it.
9 In the previous challenges, the challenge program itself (e.g., '/challenge/solve'), which runs
10 as the 'root' user (and, thus, can read the flag), read this file and printed its contents, but
11 this level is harder.
12
13 Like many of the other challenges on the platform, this challenge's '/challenge/solve' program
14 will not read the flag file directly.
15 However, it will make the flag world-readable when you run it!
16 After that, you will need to read '/flag' yourself (e.g., using 'cat /flag' or a text editor),
17 and submit its contents as the solution.
18
```

```

  WELCOME
  TO
  THE
  DOJO

Making the flag world readable (chmod 644 /flag).
.....DONE! Go read /flag (e.g., using cat)!
bash-5.2$
```



The screenshot shows a terminal window titled "Terminal - hacker@w...". The terminal shows the command `/challenge/solve` being executed. The output is a ASCII art of a person standing, followed by the text "WELCOME TO THE DOJO". Below this, it says "Making the flag world readable (chmod 644 /flag).DONE! Go read /flag (e.g., using cat)!" and then the prompt `hacker@welcome-the-flag-file:~$`.

```
hacker@welcome-the-flag-file:~$ /challenge/solve

  WELCOME
  TO
  THE
  DOJO

Making the flag world readable (chmod 644 /flag).
.....DONE! Go read /flag (e.g., using cat)!
hacker@welcome-the-flag-file:~$
```

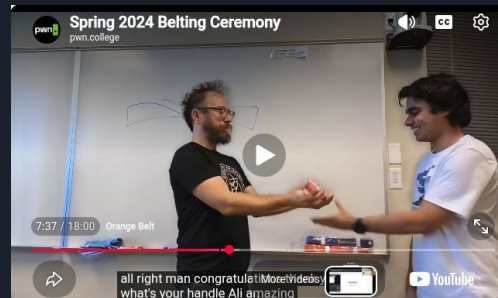
Arizona State University



- Computer Science (Cybersecurity), BS - 120 credit hours

Computer Science Lower Division
CSE 110 Principles of Programming (QTRS)
EEE 120 Digital Design Fundamentals
CSE 205 Object-Oriented Programming and Data Structures (QTRS)
CSE 230 Computer Organization and Assembly Language Programming
CSE 240 Introduction to Programming Languages
FSE 100 Introduction to Engineering
Computer Science Upper Division
CSE 301 Computing Ethics
CSE 310 Data Structures and Algorithms
CSE 330 Operating Systems
CSE 340 Principles of Programming Languages
CSE 355 Introduction to Theoretical Computer Science
CSE 360 Introduction to Software Engineering
CSE 365 Information Assurance
CSE 485 Computer Science Capstone Project I
CSE 486 Computer Science Capstone Project II
MAT 343 Applied Linear Algebra
IEE 380 Probability and Statistics for Engineering Problem Solving (QTRS)

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Cybersecurity Focus Courses
^ Upper Division Cybersecurity Focus Courses
CSE 466 Computer Systems Security
CSE 467 Data and Information Security
CSE 468 Computer Network Security
CSE 469 Computer and Network Forensics
CSE 494 Topic: Artificial Intelligence for Cyber Security
CSE 412 Database Management
OR CSE 434 Computer Networks
OR CSE 445 Distributed Software Development
^ Upper Division Cybersecurity Electives
CSE 445 Distributed Software Development
CSE 460 Software Analysis and Design
CSE 463 Introduction to Human Computer Interaction
CSE 464 Software Quality Assurance and Testing
CSE 466 Computer Systems Security
CSE 467 Data and Information Security
CSE 468 Computer Network Security
CSE 469 Computer and Network Forensics
CSE 471 Introduction to Artificial Intelligence
CSE 494 Topic: Artificial Intelligence for Cyber Security



Coursework -> Belts



- CSE 240 (Formerly) - Intro to Programming Languages Dojo
 - Intro to Programming Languages Model
 - Linux, C, C++, Scheme (functional programming)
- CSE 365 Information Assurance (500+ challenges)
 - Linux Luminarium Dojo
 - Computing 101 Dojo
 - Playing with Programs Dojo
 - Intro to Cyber Security Dojo / Orange Belt
- CSE 466 Computer Systems Security 🥷
 - Yellow, Green, and Blue belts?
 - Challenges have CPU vulnerabilities / legacy hardware
- CSE 598 Special Topics
 - Applied Vulnerability Research
 - Advanced SW Exploitation

- Much of pwn.college is integrated into ASU coursework
 - Pwn.college solve percentage is most / all of their students grade
 - See ASU academic integrity policy
-
- No write-ups
 - No solutions on Github
 - No sharing of flags (at least with ASU students)

Plagiarism and Cheating

Plagiarism or any form of cheating in assignments or projects is subject to serious academic penalty. To understand your responsibilities as a student read: [ASU Student Code of Conduct](#) and [ASU Student Academic Integrity Policy](#). There is a zero tolerance policy in this class: any violation of the academic integrity policy will result in a zero on the assignment and the violation will be reported to the Dean's office. Plagiarism is taken very seriously in this course.

Examples of academic integrity violations include (but are not limited to):

- Sharing code with a fellow student (even if it's only a few lines).
- Collaborating on code with a fellow student (unless explicitly allowed).
- Using another student's solution to solve a challenge and get a flag.
- Sharing a flag with another student (NEVER ALLOWED UNDER ANY CIRCUMSTANCES).

Posting your assignment solutions online is expressly forbidden, and will be considered a violation of the academic integrity policy. Note that this includes working out of a public Github repository. The [Github Student Developer Pack](#) provides unlimited private repositories while you are a student, making it easy to begin with a private GitHub repository.

Value of Pwn College

- Multiple engineers I've worked with have told me "Pwn.college is how I got this job"
 - Challenging / Rewarding
 - Demonstrate and understand real world offensive cyber
- Better than most cyber security certificates
 - 480 players have completed 90% of orange belt
 - 225 players have completed 90% of yellow belt
 - 210 players have completed 90% of green belt
 - 188 players have completed 90% of blue belt
- Understanding of any of these concepts will help you in any CS field



Westshore Dojo

1. Signup on pwn.college for an account
2. <https://pwn.college/dojo/westshore-cs-club~0b45ad3f/join/>



Modules

Pseudo Random
Number
Generators
(PRNG)

0 / 2

Hashing

0 / 2

Cryptography

0 / 6

Cracks and
Keygens

0 / 2

General Skills

1 / 2

Pwn

0 / 1

Coding

0 / 1

Reverse
Engineering

0 / 6

Forensics

0 / 1

Impossible

0 / 1



Attributions

- <https://cybersecurityguide.org/>
- [DEFCON31-Defending KA-SAT](#)
- <https://forum.nhl94.com/index.php?/topic/34793-nhl-94-2025-edition-by-adam-catalyst/>